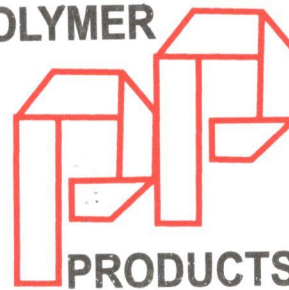


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POLYMER

ELASTOMERIC BEARING INSTALLATION AND MAINTENANCE METHODOLOGY

AS PER IRC:83-2018 (PART-II) SPECIFICATIONS

9 INSTALLATIONS: -

9.1 Care shall be taken in packing, transportation, storage and handling to avoid any mechanical damage, contamination with oil, grease and dirt, undue exposure to sunlight and weather.

9.2 Bearings shall be installed in the structure as specified or approved by the engineer to ensure that right bearing is being installed at the right location.

9.3 Bearings must be placed between true horizontal surfaces (maximum tolerance 0.2 per cent perpendicular to load) and at true plan position of their control lines marked on receiving surfaces (maximum tolerance ± 3 mm). Concrete surfaces shall be free from local irregularities (maximum tolerance ± 1 mm in height).

9.4 Departures from common planarity of twin or multiple bearings shall be within such tolerance as may be specified or approved by the engineer.

9.5 Design shall be checked for the actual inclination in seating if larger inaccuracies than those specified in Clause 9.3 are permitted.

9.6 For cast-in-place concrete construction of superstructure, where bearings are installed prior to its concreting, the forms around the bearings shall be soft enough for easy removal. Forms shall also fit the bearings snugly and prevent any leakage of mortar grout. Any mortar contaminating the bearings during concreting shall be completely removed before setting.

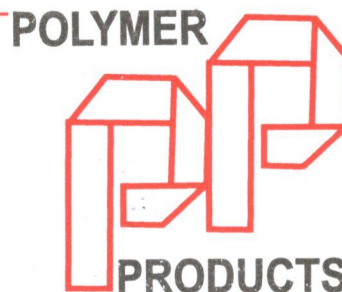
9.7 For precast concrete or steel superstructure elements, fixing of bearing to them may be done by application of epoxy resin adhesive to interface, after specified surface preparation. The specification for adhesive material, workmanship and control shall be approved by the engineer. Care shall be taken to guard against faulty application and consequent behaviour of the adhesive layer as a lubricant. The bonding by the adhesive shall be deemed effective only as a device for installation and shall not be deemed to secure bearing against displacement for purpose of design.

9.8 Lifting of a cast-in-place post-tensioned bridge deck for relieving time dependent deformation shortly after installation of bearings should be avoided. In case such lifting is



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unavoidable, the lifting arrangement, proper seating of the girder on the bearing, etc. shall be rigidly controlled to avoid any risk of misalignment.

9.9 After installation, bearing and their surrounding areas shall be left clean.

9.10 Under load, bulging of the rubber layer between the reinforcing steel laminates on free exposed perimeter is a normal phenomenon and is not detrimental to the performance of the bearing unless any crack or evidence of bond failure is detected.

10 MAINTENANCES: -

10.1 The structure should be designed and detailed in such a way that the bearings are easily accessible after installation for inspection and maintenance. Arrangements for insertion of jacks to lift the bridge deck shall be made in detailing of structure.

10.2 The bearings shall be subjected to planned maintenance care.

10.3 The exposed bearing surface shall be maintained clean and free from contamination with grease or oil, etc.

10.4 Annual routine maintenance inspection or special maintenance inspection of all bearings shall be made to check the following criteria: – The top and bottom load bearing surfaces shall be in full contact with the plinth (bottom supporting surface) and the soffit (top supporting surface). If there is imperfect contact between the surfaces stated above, the angle between the soffit and the plinth shall be checked against the design specifications. – Shear deflection is expected phenomenon in Elastomeric bearing. However, the magnitude of the shear deflection of each bearing shall be checked to ensure that it is within the design specifications. – A visual inspection shall be made of all the accessible edges. A note shall be made of the size and position of any cracks or splits, or uneven bulges. Near uniform bulging is however expected phenomenon in elastomeric bearing. – Examine the plinth and soffit for signs of displacement from bearing original position (black marks may give an indication of movement). – If applicable, examine the sliding surfaces for cleanliness and that the movements are within the design range and report results. – If applicable, examine the protective coating and/or dust protection for signs of deterioration and report results

10.5 Damaged bearings shall be replaced immediately. To avoid differences in stiffness, all adjacent bearings on the same line of support shall also be replaced.

